1 Appendix

1.1 Transformer Self-Attention as a Fully Connected Graph

This subsection formalizes the graph interpretation used in Section 3. In an unmasked Transformer self-attention layer, every token, particle, or component can attend to every other token, particle, or component. Thus the layer can be viewed as message passing on a complete directed graph with self-loops. This follows directly from scaled dot-product attention, where the output is computed by applying a softmax to all query–key dot products and then taking a weighted sum of all values [10]. This viewpoint is also consistent with graph-transformer formulations that describe the original Transformer as operating on fully connected graphs over sequence elements [3].

Let $H = [\boldsymbol{h}_1, \ldots, \boldsymbol{h}_n]^\top \in \mathbb{R}^{n\times d}$ denote the input node-feature matrix for n elements. For one attention head, define

$$Q = HW_Q, \qquad K = HW_K, \qquad V = HW_V,$$

where the ith rows are the query $\boldsymbol{q}_i$, key $\boldsymbol{k}_i$, and value $\boldsymbol{v}_i$, respectively. The attention weight from sender node j to receiver node i is

$$\alpha_{ij} = \frac{\exp\left(\boldsymbol{q}_i^\top \boldsymbol{k}_j/\sqrt{d_k}\right)}{\sum_{r=1}^{n} \exp\left(\boldsymbol{q}_i^\top \boldsymbol{k}_r/\sqrt{d_k}\right)}.$$

The receiver update is therefore

$$\boldsymbol{z}_i = \sum_{j=1}^{n} \alpha_{ij}\boldsymbol{v}_j, \qquad i = 1, \ldots, n.$$

Equivalently, in matrix form,

$$Z = AV, \qquad A = \operatorname{softmax}\left(\frac{QK^\top}{\sqrt{d_k}}\right),$$

where the softmax is applied row-wise.

Define the directed attention graph as

$$G_{\text{att}} = (\mathcal{V}, \mathcal{E}_{\text{att}}), \qquad \mathcal{V} = \{1, \ldots, n\},$$
$$\mathcal{E}_{\text{att}} = \{(j, i) : j, i \in \mathcal{V}\}.$$

For finite query–key scores and no mask, the softmax gives $\alpha_{ij} > 0$ for every ordered pair (j, i). Hence every sender j sends a message $\boldsymbol{v}_j$ to every receiver i, including itself. The adjacency support of the attention layer is therefore

$$\operatorname{supp}(A) = \mathbf{1}_n\mathbf{1}_n^\top,$$

which is the adjacency support of a fully connected directed graph with self-loops.

Multi-head attention repeats this complete-graph aggregation in parallel:

$$\text{head}_{m,i} = \sum_{j=1}^{n} \alpha_{ij}^{(m)} \boldsymbol{v}_j^{(m)},$$
$$\text{MHA}_i = \text{Concat}(\text{head}_{1,i}, \ldots, \text{head}_{M,i})W_O.$$

Thus each head learns a different dense edge-weight matrix on the same complete interaction graph, and the output projection mixes the resulting complete-graph messages. If a mask is introduced, such as a causal decoder mask or padding mask, the masked entries correspond to removed directed edges; the fully connected interpretation applies to unmasked self-attention. This observation connects the HNN-Transformer discussion in Section 3 to the over-smoothing analysis below: when dense attention maps are repeatedly applied across layers, they behave like global graph-propagation operators that can mix initially distinct node or component features.

1.2 Mathematical Proof of Oversmoothing in Graph Neural Networks

Oversmoothing is the phenomenon that, as the number of message-passing layers grows, node representations in a graph neural network become increasingly indistinguishable. The proof below shows this effect for the standard linearized graph convolutional propagation used by graph convolutional networks (GCNs) [6]. This is consistent with later asymptotic analyses showing that deep GNNs can lose node-distinguishing expressive power exponentially fast [8].

Definition 1 (Symmetric normalized propagation matrix)**.** Let $G = (V, E)$ be a connected, undirected graph with node set $V = \{1, \ldots, n\}$, where $n \geq 2$; the one-node case is immediate. Let $A \in \mathbb{R}^{n\times n}$ be the adjacency matrix of G, where $A_{ij} = 1$ if $\{i, j\} \in E$ and $A_{ij} = 0$ otherwise. Let I_n denote the $n \times n$ identity matrix, and let $\mathbf{1} \in \mathbb{R}^n$ denote the all-ones column vector. Following the usual GCN convention, add one self-loop to every node and define

$$\widehat{A} = A + I_n,$$

$$\widehat{d}_i = \sum_{j=1}^{n} \widehat{A}_{ij}, \qquad \widehat{D} = \operatorname{diag}(\widehat{d}_1, \ldots, \widehat{d}_n).$$

Here $\widehat{d}_i$ is the degree of node i after self-loops have been added, so $\widehat{d}_i \geq 1$ and the inverse powers of $\widehat{D}$ below are well-defined.

The symmetric normalized propagation matrix is

$$S = \widehat{D}^{-1/2}\widehat{A}\widehat{D}^{-1/2}.$$

Definition 2 (L-layer graph neural network)**.** Starting from an input feature matrix $X \in \mathbb{R}^{n\times p_0}$, let $X_i \in \mathbb{R}^{1\times p_0}$ denote its ith row. We write

$$H^{(0)} = X$$

for the initial node-feature matrix. For a linearized L-layer GCN, let $H^{(\ell)} \in \mathbb{R}^{n\times p_\ell}$ be the node-feature matrix at layer ℓ, where the ith row $H_i^{(\ell)}$ is the feature vector of node i after ℓ propagation steps. Without nonlinearities, the layer update is

$$H^{(\ell)} = SH^{(\ell-1)}W^{(\ell)}, \qquad \ell = 1, \ldots, L,$$

where $W^{(\ell)} \in \mathbb{R}^{p_{\ell-1}\times p_\ell}$ is the learnable weight matrix at layer ℓ. Define the product of layer weights by

$$W_{1:L} = W^{(1)}W^{(2)}\cdots W^{(L)} \in \mathbb{R}^{p_0\times p_L}.$$

Iterating the recursion gives

$$H^{(L)} = S^L X W_{1:L}.$$

Thus $H^{(L)}$ is the node-feature matrix at layer L. The question is what happens to $H^{(L)}$ as L becomes large.

We next recall the graph and matrix conditions used in the convergence argument.

Definition 3 (Strongly connected directed graph)**.** Let $G = (V, E)$ be a directed graph. A directed path from u to v is a sequence of vertices

$$u = x_0, x_1, \dots, x_m = v$$

such that $(x_{r-1}, x_r) \in E$ for each $r = 1, \dots, m$. The directed graph G is *strongly connected* if, for every ordered pair of vertices $u, v \in V$, there exists a directed path from u to v. Equivalently, every vertex is reachable from every other vertex while respecting edge directions.

The definitions below follow the standard finite-state Markov-chain and nonnegative-matrix terminology [2, 7].

Definition 4 (Stochastic, irreducible, aperiodic, and primitive matrices)**.** Let $P \in \mathbb{R}^{n\times n}$ be a nonnegative matrix, meaning $P_{ij} \geq 0$ for all i, j.

(a) P is *stochastic* if every row sums to one:

$$\sum_{j=1}^{n} P_{ij} = 1, \qquad i = 1, \dots, n.$$

(b) P is *irreducible* if every state can reach every other state: for all $i, j \in \{1, \dots, n\}$, there exists an integer $k \geq 0$ such that

$$(P^k)_{ij} > 0.$$

Equivalently, the directed graph with edge $i \longrightarrow j$ whenever $P_{ij} > 0$ is strongly connected.

(c) For an irreducible P, the period of state i is

$$\mathrm{period}(i) = \gcd\{k \geq 1 : (P^k)_{ii} > 0\}.$$

where gcd is the greatest common divisor. The matrix P is *aperiodic* if $\mathrm{period}(i) = 1$ for every state i; in the irreducible case, all states have the same period.

(d) P is *primitive* if some positive power of P is entrywise positive: there exists $m \geq 1$ such that

$$(P^m)_{ij} > 0, \qquad \text{for all } i, j.$$

For finite nonnegative matrices, primitivity is equivalent to irreducibility plus aperiodicity.

The key spectral tool is the following Perron–Frobenius theorem for primitive stochastic matrices.

Lemma 1 (Perron–Frobenius theorem [4, 9]). *Let $P \in \mathbb{R}^{n\times n}$ be a primitive stochastic matrix. Let its eigenvalues be $\lambda_1 = 1, \lambda_2, \dots, \lambda_n$. Then the following statements hold:*

(a) *The eigenvalue 1 is simple and dominant. In particular, all other eigenvalues lie strictly inside the unit disk:*

$$|\lambda_j| < 1, \qquad j = 2, \dots, n.$$

(b) *There exists a unique eigenvector $\boldsymbol{v} \in \mathbb{R}^n$ associated with the eigenvalue 1, such that*

$$\boldsymbol{v}^\top P = \boldsymbol{v}^\top, \qquad \boldsymbol{v}^\top \mathbf{1} = 1.$$

(c) *This unique eigenvector $\boldsymbol{v}$ has all positive entries:*

$$v_i > 0, \qquad i = 1, \dots, n.$$

To measure convergence of Markov-chain distributions, we use total variation distance.

Definition 5 (Total variation distance [7])**.** Let μ and ν be probability distributions on a finite state space $\mathcal{X}$. The total variation distance between μ and ν is

$$\|\mu - \nu\|_{\mathrm{TV}} = \frac{1}{2} \sum_{x\in\mathcal{X}} |\mu(x) - \nu(x)|.$$

Lemma 2 (Convergence theorem for Markov chains [7]). *Let P be an irreducible and aperiodic stochastic matrix with stationary distribution $\boldsymbol{v}$ satisfying $\boldsymbol{v}^\top P = \boldsymbol{v}^\top$ and $\boldsymbol{v}^\top \mathbf{1} = 1$. Then P converges to its stationary distribution: for every state $i \in \mathcal{X}$,*

$$\lim_{L\to\infty} \|P^L(i, .) - \boldsymbol{v}^\top\|_{\mathrm{TV}} = 0.$$

Equivalently,

$$\lim_{L\to\infty} P^L(i, j) = v_j, \qquad i, j \in \mathcal{X}.$$

Therefore, for any initial row distribution $\boldsymbol{\alpha}^\top$,

$$\lim_{L\to\infty} \boldsymbol{\alpha}^\top P^L = \boldsymbol{v}^\top.$$

Lemma 3 (Spectral theorem for real symmetric matrices [5]). *Let $P \in \mathbb{R}^{n\times n}$ be symmetric. Then all eigenvalues of P are real and there exists an orthonormal eigenbasis $\{\boldsymbol{u}_1, \dots, \boldsymbol{u}_n\}$ of $\mathbb{R}^n$ with corresponding eigenvalues $\mu_1, \dots, \mu_n$. Equivalently,*

$$P = \sum_{j=1}^{n} \mu_j \boldsymbol{u}_j \boldsymbol{u}_j^\top.$$

Consequently, for every integer $L \geq 0$,

$$P^L = \sum_{j=1}^{n} \mu_j^L \boldsymbol{u}_j \boldsymbol{u}_j^\top.$$

Theorem 1 (Convergence of linearized L-layer GCN node features). *Let S be the symmetric normalized propagation matrix defined above, and let*

$$\boldsymbol{v} = \frac{\widehat{D}^{1/2}\mathbf{1}}{\|\widehat{D}^{1/2}\mathbf{1}\|_2}$$

be the unit eigenvector of S associated with eigenvalue 1. For any matrix $Y \in \mathbb{R}^{n\times p}$,

$$\lim_{L\to\infty} S^L Y = \boldsymbol{v}\boldsymbol{v}^\top Y.$$

Thus repeated propagation drives node features into the one-dimensional eigenspace associated with eigenvalue 1.

Proof. Because $G = (V, E)$ is connected and one self-loop is added to every node, the random-walk matrix

$$P = \widehat{D}^{-1}\widehat{A}$$

is stochastic, irreducible, and aperiodic. By Lemma 2, P converges to its unique stationary distribution; equivalently, by Lemma 1, the eigenvalue 1 is simple and all other eigenvalues have modulus strictly smaller than 1.

The matrices P and S are similar because

$$S = \widehat{D}^{1/2} P \widehat{D}^{-1/2}.$$

Therefore S has the same eigenvalues as P. Moreover, $S\boldsymbol{v} = \boldsymbol{v}$ for

$$\boldsymbol{v} = \frac{\widehat{D}^{1/2}\mathbf{1}}{\|\widehat{D}^{1/2}\mathbf{1}\|_2}.$$

Since S is symmetric, Lemma 3 gives an orthonormal eigenbasis for S and the spectral expansion of its powers. Because the eigenvalue 1 is simple, we may take $\boldsymbol{u}_1 = \boldsymbol{v}$ and $\lambda_1 = 1$, so

$$S^L = \boldsymbol{v}\boldsymbol{v}^\top + \sum_{j=2}^{n} \lambda_j^L \boldsymbol{u}_j \boldsymbol{u}_j^\top, \qquad |\lambda_j| < 1.$$

Thus $S^L \to \boldsymbol{v}\boldsymbol{v}^\top$ as $L \to \infty$. Multiplying by any matrix $Y \in \mathbb{R}^{n\times p}$ gives

$$\lim_{L\to\infty} S^L Y = \boldsymbol{v}\boldsymbol{v}^\top Y.$$

□

1.3 Working Example for Theorem 1

Consider the graph with adjacency matrix

$$A = \begin{bmatrix} 0 & 1 & 1 & 1 & 0 \\ 1 & 0 & 1 & 0 & 0 \\ 1 & 1 & 0 & 1 & 0 \\ 1 & 0 & 1 & 0 & 1 \\ 0 & 0 & 0 & 1 & 0 \end{bmatrix}$$

with initial node feature

$$\boldsymbol{x} = \begin{bmatrix} 0.1 \\ 0.3 \\ 0.2 \\ 0 \\ 0.4 \end{bmatrix}.$$

After adding self-loops, $\widehat{A} = A + I_5$ and

$$\widehat{D} = \mathrm{diag}(4, 3, 4, 4, 2).$$

The symmetric normalized propagation matrix $S = \widehat{D}^{-1/2}\widehat{A}\widehat{D}^{-1/2}$ is

$$S \approx \begin{bmatrix} 0.2500 & 0.2887 & 0.2500 & 0.2500 & 0 \\ 0.2887 & 0.3333 & 0.2887 & 0 & 0 \\ 0.2500 & 0.2887 & 0.2500 & 0.2500 & 0 \\ 0.2500 & 0 & 0.2500 & 0.2500 & 0.3536 \\ 0 & 0 & 0 & 0.3536 & 0.5000 \end{bmatrix}.$$

The eigenvector associated with eigenvalue 1 is

$$\boldsymbol{v} = \frac{1}{\sqrt{17}} \begin{bmatrix} 2 \\ \sqrt{3} \\ 2 \\ 2 \\ \sqrt{2} \end{bmatrix}.$$

Numerically,

$$\boldsymbol{v} \approx \begin{bmatrix} 0.4851 \\ 0.4201 \\ 0.4851 \\ 0.4851 \\ 0.3430 \end{bmatrix}.$$

Using identity layer weights and the scalar-feature propagation $\boldsymbol{h}^{(L)} = S^L\boldsymbol{x}$, the limiting vector is

$$\boldsymbol{h}^\star = \boldsymbol{v}\boldsymbol{v}^\top\boldsymbol{x}.$$

Numerically,

$$\boldsymbol{h}^\star \approx \begin{bmatrix} 0.1983 \\ 0.1717 \\ 0.1983 \\ 0.1983 \\ 0.1402 \end{bmatrix}.$$

the simulation result for the Theorem 1 shown in the Table 1

1.4 Definition of sender R_s and receiver R_r relation matrices

This subsection gives the formal definition of the sender and receiver relation matrices used in Section 2.3 and Eq. 4. Following the relation-indexing convention used in GNN [1], consider a directed graph with n nodes and m directed edges. Let the jth edge be written as

$$e_j = (s_j, r_j), \qquad j = 1, \ldots, m,$$

where $s_j \in \{1, \ldots, n\}$ is the sender node and $r_j \in \{1, \ldots, n\}$ is the receiver node. The sender relation matrix $R_s \in \{0,1\}^{n\times m}$ and receiver relation matrix $R_r \in \{0,1\}^{n\times m}$ are defined column-wise by

$$(R_s)_{ij} = \begin{cases} 1, & i = s_j, \\ 0, & i \neq s_j, \end{cases}$$

$$(R_r)_{ij} = \begin{cases} 1, & i = r_j, \\ 0, & i \neq r_j. \end{cases}$$

Thus each column of R_s and each column of R_r is a one-hot vector identifying the sender or receiver of one directed edge:

$$\sum_{i=1}^{n} (R_s)_{ij} = 1, \qquad \sum_{i=1}^{n} (R_r)_{ij} = 1,$$
$$j = 1, \ldots, m.$$

The row sums of R_s count outgoing edges from each node, while the row sums of R_r count incoming edges to each node.

Let $X \in \mathbb{R}^{n\times d}$ denote a node-state matrix, where the ith row $\boldsymbol{x}_i$ is the feature vector of node i. Then

$$(R_s^\top X)_j = \boldsymbol{x}_{s_j}, \qquad (R_r^\top X)_j = \boldsymbol{x}_{r_j},$$

so $R_s^\top X$ collects sender-node features for all edges and $R_r^\top X$ collects receiver-node features for all edges. After an edge model computes edge messages $M \in \mathbb{R}^{m\times d_m}$, multiplication by R_r aggregates incoming messages at receiver nodes:

$$(R_r M)_i = \sum_{j: r_j = i} M_j.$$

This is the role of $R_r\phi_{\theta_1}(R_r^T(\boldsymbol{p},\boldsymbol{q}); R_s^T(\boldsymbol{p},\boldsymbol{q}); E)$ in Eq. 4: the edge network first forms directed interaction messages from receiver features, sender features, and edge attributes, and R_r then sums the messages incident to each receiving node.

For example, if the graph is constructed as in Figure 1 with three nodes and all directed pairwise edges excluding self-loops, then $n = 3$ and $m = 6$. Using the edge ordering

$$e_1 = (B, A), \qquad e_2 = (A, B), \qquad e_3 = (B, C),$$
$$e_4 = (C, B), \qquad e_5 = (C, A), \qquad e_6 = (A, C),$$

the relation matrices R_s and R_r are as follows:

L	$h_1^{(L)}$	$h_2^{(L)}$	$h_3^{(L)}$	$h_4^{(L)}$	$h_5^{(L)}$	$\|\boldsymbol{h}^{(L)} - \boldsymbol{h}^\star\|_2$
0	0.100000	0.300000	0.200000	0.000000	0.400000	0.364592
1	0.161603	0.186603	0.161603	0.216421	0.200000	0.082563
2	0.188774	0.155502	0.188774	0.205617	0.176517	0.042614
3	0.190681	0.160823	0.190681	0.208199	0.160955	0.027624
5	0.195237	0.166465	0.195237	0.202460	0.149273	0.012074
10	0.197890	0.171038	0.197890	0.198798	0.141351	0.001532
20	0.198265	0.171697	0.198265	0.198279	0.140217	0.000025
50	0.198271	0.171707	0.198271	0.198271	0.140199	0.000000

Table 1: Simulation of the working example for Theorem 1. The iterates $\boldsymbol{h}^{(L)} = S^L\boldsymbol{x}$ converge to $\boldsymbol{h}^\star$, a scalar multiple of the eigenvector v associated with eigenvalue 1.

$$R_s = \begin{array}{c} \\ \text{nodeA} \\ \text{nodeB} \\ \text{nodeC} \\ \\ \end{array} \begin{array}{c} \begin{array}{cccccc} \text{edge}_1 & \text{edge}_2 & \text{edge}_3 & \text{edge}_4 & \text{edge}_5 & \text{edge}_6 \end{array} \\ \begin{bmatrix} 0 & 1 & 0 & 0 & 0 & 1 \\ 1 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 1 & 0 \end{bmatrix} \\ \begin{array}{cccccc} =1 & =1 & =1 & =1 & =1 & =1 \end{array} \end{array} \begin{array}{c} \\ =2 \\ =2 \\ =2 \\ \\ \end{array}$$

$$R_r = \begin{array}{c} \\ \text{nodeA} \\ \text{nodeB} \\ \text{nodeC} \\ \\ \end{array} \begin{array}{c} \begin{array}{cccccc} \text{edge}_1 & \text{edge}_2 & \text{edge}_3 & \text{edge}_4 & \text{edge}_5 & \text{edge}_6 \end{array} \\ \begin{bmatrix} 1 & 0 & 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 1 \end{bmatrix} \\ \begin{array}{cccccc} =1 & =1 & =1 & =1 & =1 & =1 \end{array} \end{array} \begin{array}{c} \\ =2 \\ =2 \\ =2 \\ \\ \end{array}$$

In this example, every column contains exactly one nonzero entry because each directed edge has one sender and one receiver. The row sums are all equal to 2 because, in the complete directed three-node graph without self-loops, each node sends messages to two other nodes and receives messages from two other nodes.

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